Kevin Benny Mathew

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Education

2023 - 2027	B.E. (Hons.) Electronics and Communication Engineering - BITS Pilani, Goa Campus	**8.21/10** (Upto Semester 4)
May 2023	ISC (Class XII) - The Bishop's School, Camp	**97.6%**
May 2021	ICSE (Class X) - St. Mary's School, Pune	**99.0%**

Skills & Proficiency

Languages	Verilog, C, C++, Python, Java
Key Courses	VLSI Design, Digital Design, Microprocessors & Interfacing, Digital Signal Processing, Control Systems, Communication Systems, Operating Systems (Ongoing), Computer Architecture (Ongoing)
Technologies	Microcontrollers (ESP32, Teensy, ESP8266, Arduino), ROS/ROS2, Linux, Git
Tools	KiCAD, Vivado, Cadence Virtuoso, PlatformIO, MATLAB, Fusion 360, Solidworks, Onshape, Android Studio

Work Experience

Summer Intern May 2025 - Aug 2025

- Worked at the **Indian Meteorological Department** on hardware-software integration and algorithm development.
- Developed embedded firmware for ESP8266 hardware using C++ and PlatformIO.
- Implemented custom fuzzy logic and Kalman filters for high-accuracy sensor data acquisition.
- Conducted data analysis using Python and documented results for research publication.

Projects

Project Kratos: Mars Rover (Electronics Subsystem) Aug 2024 - Present

- Redesigned the electronics hardware architecture for modularity, easier assembly, and improved debugging.
- Migrated the control stack from ROS Noetic to ROS2 Humble.
- Designed custom mounts for cameras and sensors using Fusion 360.

Low Cost Electromechanical Tensile Tester May 2025 - Dec 2025

- Executed end-to-end development including electronics architecture, mechanical design, and 3D modeling.
- Integrated motion control systems using Teensy microcontrollers, stepper motors, and a custom UI.
- Developed a horizontal tensile tester for the characterization of soft materials under Dr. Tushar Sakorikar.

Project Vulcan: Humanoid Robot Feb 2024 - Present

- Implemented audio amplification circuits for microphones located in the robot's ears.
- Led the bionic arm subsystem, developing a 7-DOF biomimetic robotic arm emulating human movement.
- Improved internal mechanical systems for facial movements, specifically eyes and mouth.

Development of Flexible Tactile Sensors Dec 2025 - Present

– Developing tactile sensors for robotic perception under Dr. Tushar Sakorikar.

– Project is currently in literature review and ideation phase with plans to start fabrication by February 2026.

Leadership & Volunteering

Research Head - Electronics and Robotics Club (ERC), BITS Goa Apr 2025 - Present

- Guiding **75+** students across multiple robotics projects (Vulcan, Snakebot, Quadruped)
- Managing project progress and integrating electronics, mechanical, and automation verticals.
- Organized and designed ROS2-based hackathon with **200+ participants nationwide**

Committee Member - Sandbox Innovation Lab, BITS Goa Feb 2024 - Present

Maintaining lab equipment of the campus student-run (3D printers, laser cutters) and managing tech club purchases.

Volunteer - Abhigyaan, BITS Goa Oct 2023 - Present

Provided evening tuition classes for K-12 children of campus non-teaching staff.

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